Companion to Bird & de Moor, Algebra of Programming

§5. Datatypes in Allegories

§5.1. Relators

(5.1a)

A **relator** `F : 𝒞→𝒟` is a 2-functor between allegories: it preserves `𝟙`, composition and `⊑`.

(5.1b)

	the statement
1	`Map(f) ⟹ Map(F(f))` and `Map(f) ⟹ F(f°) = F(f)°` . Lemma 5.1
2	Over a **tabular** allegory a functor is a relator ⟺ it preserves `°`. Theorem 5.1 the book's reverse direction (preserving `°` ⟹ relator) needs the source to be `Rel`: the counterexample over a tabular allegory that is not `Rel` shows it can fail
3	`F(R°) = F(R)°` for every `R`, so `F(R)°` needs no bracket.
4	Two relators agreeing on maps are equal. Corollary 5.1
5	`Coreflexive(X) ∧ Coreflexive(Y) ⟹ F(X ∩ Y) = F(X) ∩ F(Y)` Ex 5.2
6	`F(R ∩ S) ⊑ F(R) ∩ F(S)` , and strictly. Ex 5.2
7	`F(dom(R)) = dom(F(R))` Ex 5.5

The **power relator** `P` — `xs P(R) ys⟺(∀a∈xs. ∃b∈ys. a R b)∧(∀b∈ys. ∃a∈xs. a R b)` — is where the fourth is strict: for `R={(a₁,b₁),(a₂,b₂)}` and `S={(a₁,b₂),(a₂,b₁)}` the pair `({a₁,a₂},{b₁,b₂})` is in `P(R)∩P(S)`, while `R∩S=∅`.

§5.2. Relational products

(5.1)

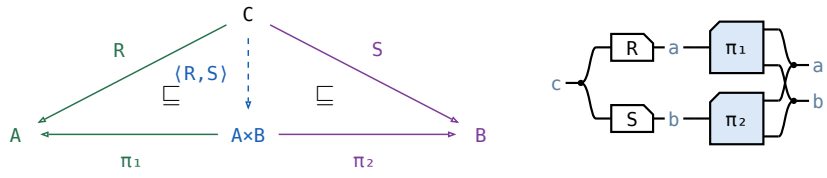
$\langle R,S\rangle \triangleq (R\ \pi_1{}^\circ) \cap (S\ \pi_2{}^\circ)$ — the **fork** of $R : C\to A$ and $S : C\to B$, with (π_1,π_2) the tabulation of $\top$.

$\langle R,S\rangle \triangleq R\pi_1{}^\circ \cap S\pi_2{}^\circ$ $c\ \langle R,S\rangle\ (a,b) \Longleftrightarrow c\ R\ a \wedge c\ S\ b$.

(5.2a)

$\langle R,S\rangle\ \pi_1 = \mathrm{dom}(S)\ R$ (5.6) $\langle R,S\rangle\ \pi_2 = \mathrm{dom}(R)\ S$ (5.7)

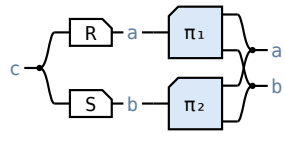
(5.2b)



A domain is coreflexive, so $\langle R,S\rangle\pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f,g\rangle$ is unique. In Rel, $c\ \langle R,S\rangle\ (a,b)$ iff $c\ R\ a$ and $c\ S\ b$ — copy c, then R on one strand and S on the other, which is $\lhd(R\otimes S)$ on the right.

No $^\circ$ survives the translation. $\pi_1 = \mathbb{1}\otimes\multimap$ discards the second component, so $\pi_1{}^\circ = \mathbb{1}\otimes\multimap$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(5.2c)



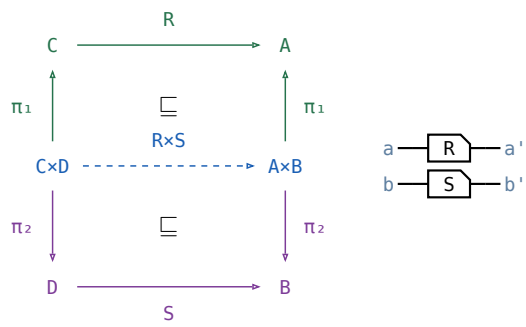
$\multimap\!\rhd = \mathbb{1}$ on each half

§5.2.1. Relational product R×S

(5.2)

$R \times S \triangleq \langle \pi_1\ R, \pi_2\ S\rangle$ — the relational product $R\times S \triangleq \langle \pi_1 R, \pi_2 S\rangle$.

(5.2.1a)

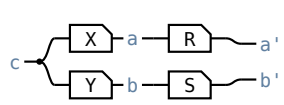


Right-then-up is $(R\times S)\pi_1$, up-then-right is $\pi_1 R$, and $(R\times S)\pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In Rel, $(c,d)\ (R\times S)\ (a,b)$ iff $c\ R\ a$ and $d\ S\ b$ — two strands side by side, no copy dot: $R\times S = R\otimes S$.

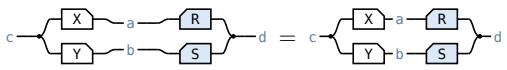
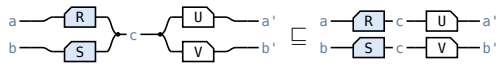
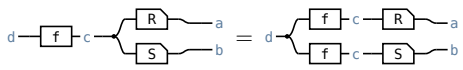
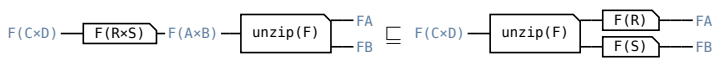
§5.2.2. Absorption

For $X : E\to C$ and $Y : E\to D$, $\langle X,Y\rangle(R\times S) = \langle XR, YS\rangle$. Both sides are this picture:

(5.3)

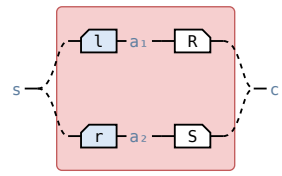
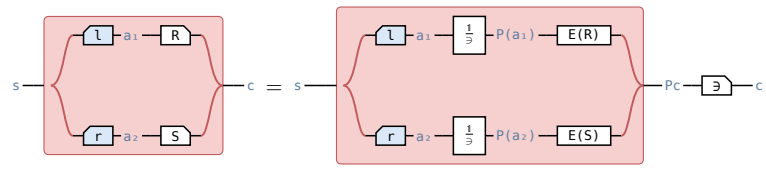
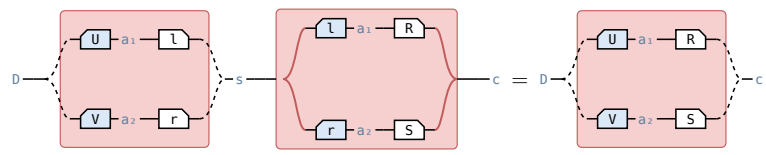


(5.2.2a)

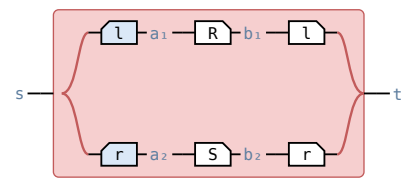
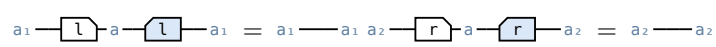
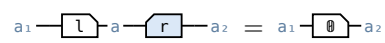
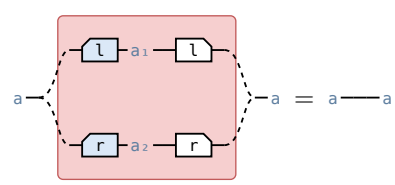
	the law	picture
1	⟨X,Y⟩ ⟨R,S⟩° = (X R°) ∩ (Y S°)	
2	⟨R,S⟩° ⟨U,V⟩ ⊑ (R° U) × (S° V)	
3	Map(f) ⟹ f ⟨R,S⟩ = ⟨f R,f S⟩ f a map; it fails for an arbitrary arrow;	
4	X ⟨R,S⟩ ⊑ ⟨X R,X S⟩ the half of the row above that holds for every X	
5	F(R × S) unzip(F) ⊑ unzip(F) (F(R) × F(S)) unzip(F)≜⟨F(π₁),F(π₂)⟩; only ⊑ for an arbitrary R, S	
6	g=curry(f)⟺(g×𝟙)eval=f reading × as the relational product, does Rel have exponentials?	

§5.3. Relational coproducts

(5.3a)

	the statement R : a₁→c, S : a₂→c	picture
1	[R,S] ≜ (l° R) ∪ (r° S) (5.9) The tape is the union — a particle entering at A+B takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
2	[R,S] = [R/∋, S/∋] ∋	
3	l [R,S] = R , r [R,S] = S l T = R ∧ r T = S ⟹ T = [R,S] ,	
4	U : a₁→D, V : a₂→D [U,V]° [R,S] = (U° R) ∪ (V° S) (5.11)	
5	Z : c→D [R,S] Z = [R Z,S Z]	

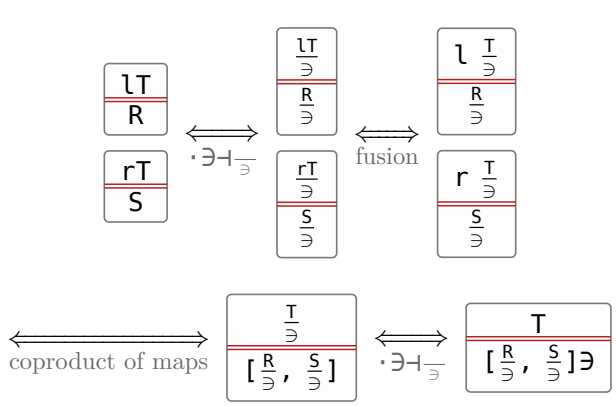
(5.3b)

	the statement R : a₁→b₁, S : a₂→b₂	picture
1	R + S ≜ [R l,S r] (5.10)	
2	l l° = 𝟙 , r r° = 𝟙 Ex 5.12	
3	l r° = 𝟘 , r l° = 𝟘 Ex 5.12	
4	(l° l) ∪ (r° r) = 𝟙 Ex 5.12	

§5.3.1. [R,S] = [R/∋, S/∋] ∋ — a case of relations is the case of their transposes, then ∋

The universal-property row is not free: l,r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\frac{\ }{\ni}$ sends an arrow A→C to a map A→PC reversibly, so the map coproduct can be applied underneath it. For any T : A+B→C,

(5.3.1a)

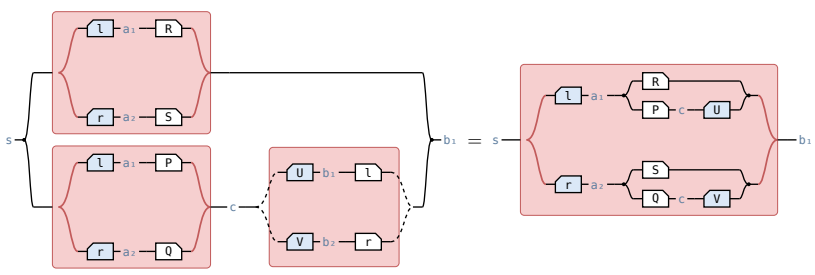


(5.3.1b)

$$\begin{tikzcd}
 & & A \arrow[dl, "\mathtt{l}"'] \arrow[dr, "\frac{R}{\ni}"] \arrow[drr, bend left, "R"] & & \\
A+B \arrow[rrr, dashed, "{[\frac{R}{\ni},\frac{S}{\ni}]}"] & & & PC \arrow[r, "\ni"] & C \\
 & & B \arrow[ul, "\mathtt{r}"] \arrow[ur, "\frac{S}{\ni}"'] \arrow[urr, bend right, "S"'] & &
\end{tikzcd}$$

Nothing here holds only up to ⊑: every triangle commutes on the nose, which is the difference from the fork above. The border spells `[R,S]=[`$\frac{R}{\ni}$`, `$\frac{S}{\ni}$`]∋`, and pushing ∋ into the union that `[·,·]` on maps already is turns that back into the definition, `(l°`$\frac{R}{\ni}$` ∪r°`$\frac{S}{\ni}$`)∋=l°R ∪ r°S`.

(5.3.1c)

	the law	picture
1	prove (5.11), and say why duality does not carry it over from the product law	
2	`(R + S) ∩ ([P,Q] [U,V]°) = (R ∩ (P U°)) + (S ∩ (Q V°))` `[P,Q][U,V]°` is a full 2×2 of composites; `R+S` is diagonal, so the meet cuts the two off-diagonal branches	

§5.4. The power relator P(R)

(5.4a)

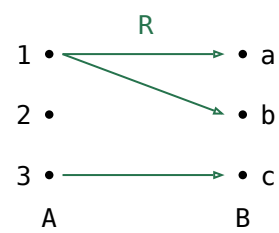
P(R) ≜ (∈ \ (R ∈)) ∩ ((∋ R) / ∋) — the power relator P(R) : PA→PB, R lifted to sets.

xs P(R) ys⟺(∀a∈xs. ∃b∈ys. a R b)∧(∀b∈ys. ∃a∈xs. a R b)

(5.4b)

	in Rel	in words	im(xs)	partner
1 (∋R)/∋	∀y∈ys. ∃x∈xs. x R y	∀y. some xs R y	ys⊆im(xs)	every y has a partner
2 (∋/(∋R))°	∀y. (∃x∈xs. x R y)→y∈ys	every y with some xs R y is in ys	ys⊇im(xs)	ys leaves out no partner
3 E(R)	ys={y|∃x∈xs. x R y}	ys = every y with some xs R y	ys=im(xs)	every y has a partner, and none is left out
4 ((∋R°)/∋)°	∀x∈xs. ∃y∈ys. x R y	∀x. x R some ys	—	every x has a partner
5 P(R)	P(R)(xs,ys) ⟺ (∀y. y ∈ ys ⟹ (∃x. x ∈ xs ∧ R(x,y))) ∧ (∀x. x ∈ xs ⟹ (∃y. y ∈ ys ∧ R(x,y)))	∀y. some xs R y and ∀x. x R some ys	—	every x and every y has a partner

(5.4c)



	xs	(∋R)/∋	(∋/(∋R))°	E(R)	((∋R°)/∋)°	P(R)
1	∅	∅	all 8 subsets of abc	∅	all 8 subsets of abc	∅
2	{1}	∅, a, b, ab	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
3	{2}	∅	all 8 subsets of abc	∅	none	none
4	{3}	∅, c	c, ac, bc, abc	c	c, ac, bc, abc	c
5	{1,2}	∅, a, b, ab	ab, abc	ab	none	none
6	{1,3}	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
7	{2,3}	∅, c	c, ac, bc, abc	c	none	none
8	{1,2,3}	all 8 subsets of abc	abc	abc	none	none

(5.4d)

	law	the reading
1	X ⊑ P(R) ⟺ X ∋ ⊑ ∋ R ∧ X° ∋ ⊑ ∋ R°	One containment, and the same one at R° — which is the definition read off the two divisions. Hence P(R°)=P(R)°, and R⊑S⟹P(R)⊑P(S) .
2	P(𝟙) = 𝟙	The straightness axiom verbatim: extensionality **is** P's unit law.
3	Map(f) ⟹ P(f) = E(f)	In Rel, xs P(f) ys⟺ys={f(a)|a∈xs}. The half at f° says every a∈xs has its f(a) on ys; f has just the one image per a, so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
4	P(R S) = P(R) P(S)	⊒ is the division cancellation laws. ⊑ is the one law in this section that is not a calculation: it needs a tabulation of P(RS).

§5.4.1. Type relators

(5.4.1a)

`T(R) = ⦇F(R,𝟙) α⦈` — the type functor `T` acts on an arrow `R : A→B` by rebuilding with `α`, applying `R` to the parameter.

(5.4.1b)

	name	law	what it says
1	the defining equation	`T(R) = ⦇F(R,𝟙) α⦈`	Rebuild the structure with `α`, applying `R` to the parameter on the way.
2	type relator	`T(R)° = T(R°)`	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

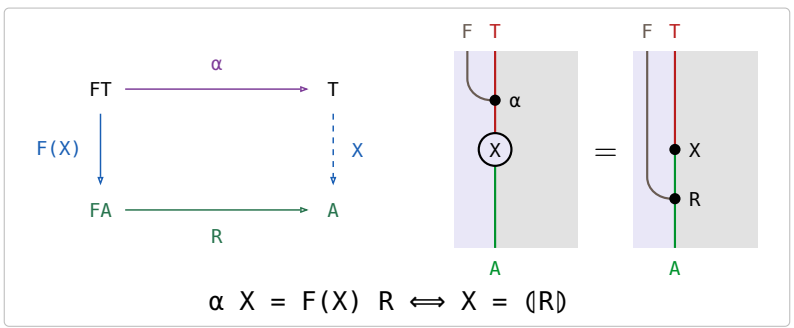
§5.5. Relational catamorphisms

(5.5a)

The initial algebra `α : F(T)→T` among the maps is also initial in the allegory:

§5.5.1. The defining equation

(5.5.1a)



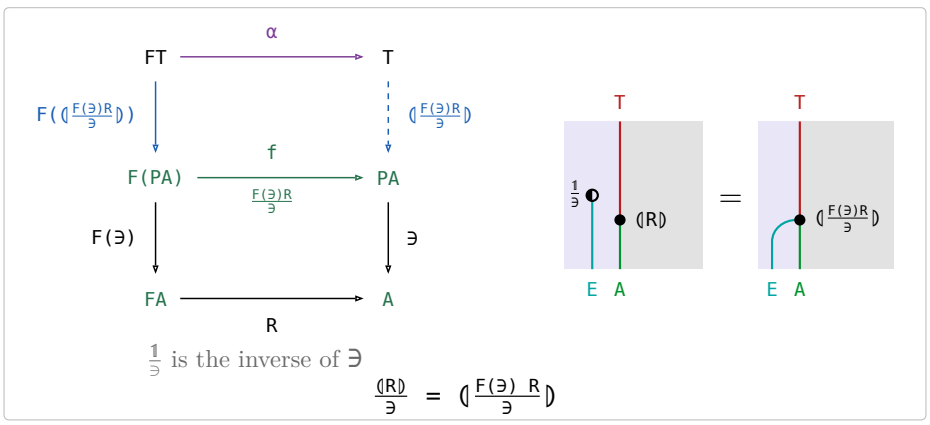
An F-algebra is a **weakened** natural transformation. A transformation `F⇒𝟙` on $\mathcal{A}$ would need a component `FX→X` at every object and a commuting square at every arrow, but F-Algebra only need it works on T and A.

(5.5.1b)

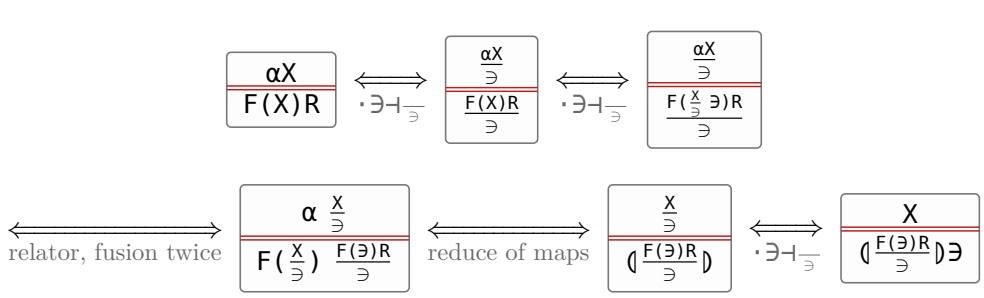
	⦇[c,f]⦈=reduce(c,f) catamorphism: arrows of the form ⦇f⦈ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing		
1	**part**	**Nat**	**[A]**
2	datatype	Nat::=zero|succ Nat	[A]::=nil|cons(A,[A])
3	base functor F	F(X)=1+X	F(X)=1+A×X
4	initial algebra α	α=[zero,succ] : 1+Nat→Nat	α=[nil,cons] : 1+A×[A]→[A]
5	the fold, pointwise	⦇[c,f]⦈(zero)=c ⦇[c,f]⦈(succ(n))=f(⦇[c,f]⦈(n))	⦇[c,f]⦈(nil)=c ⦇[c,f]⦈(cons(a,x))=f(a,⦇[c,f]⦈(x))

§5.5.2. ⦇R⦈ = ⦇$\frac{F(\ni)\ R}{\ni}$⦈ ∋ — folding with R is a fold of maps into sets, then ∋

(5.5.2a)



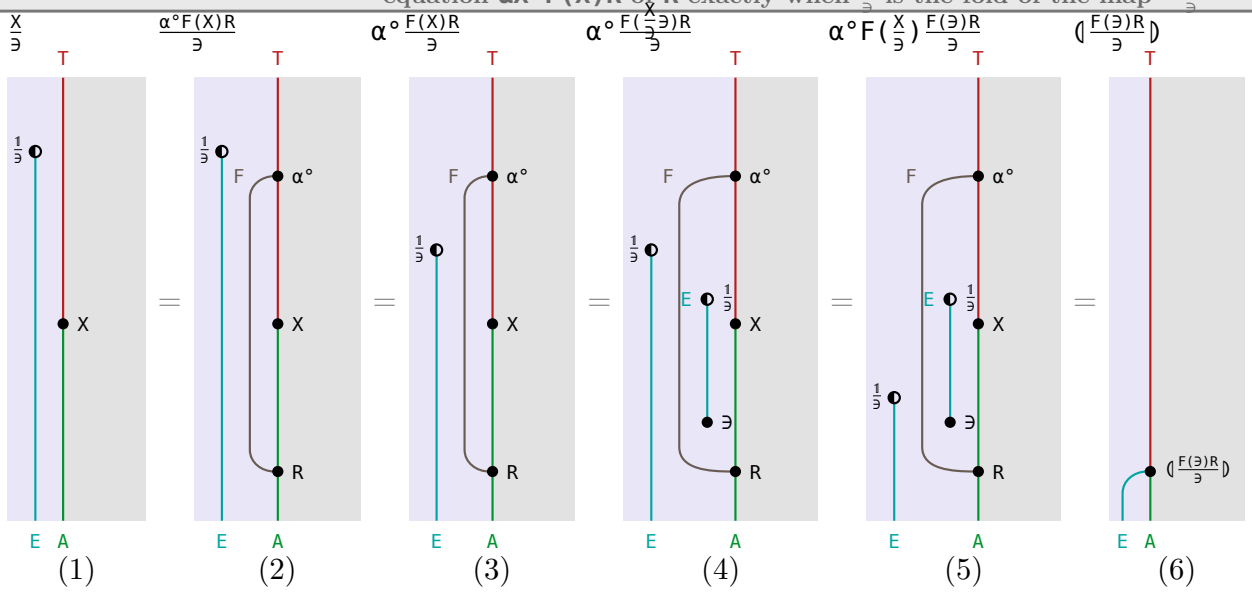
(5.12)



the last two rows at `X:=⦇R⦈`:

$$\alpha\ X = F(X)\ R \iff \frac{X}{\ni} = (\!|\frac{F(\ni)\ R}{\ni}|\!) \tag{5.5.2b}$$

in a tabular allegory, with F a relator and $\alpha : F(T) \to T$ its initial algebra: a relation X out of T satisfies the fold equation $\alpha X = F(X)R$ of R exactly when $\frac{X}{\ni}$ is the fold of the map $\frac{F(\ni)R}{\ni}$



1	$\frac{X}{\ni}$	α iso; $\alpha X = F(X)R$
2	$= \frac{\alpha^\circ\ F(X)\ R}{\ni}$	α° is a map because α is an iso, so it leaves Λ: $\frac{\alpha^\circ F(X)R}{\ni} = \alpha^\circ \frac{F(X)R}{\ni}$ — $Map(f)$ $\implies \frac{f\ R}{\ni} = f\ \frac{R}{\ni}$
3	$= \alpha^\circ\ \frac{F(X)\ R}{\ni}$	$X = \frac{X}{\ni}\ni$ — $\frac{R}{\ni}\ \ni = R$
4	$= \alpha^\circ\ \frac{F(\frac{X}{\ni}\ \ni)\ R}{\ni}$	$F(\frac{X}{\ni}\ni) = F(\frac{X}{\ni})F(\ni)$ as F is a functor, and $F(\frac{X}{\ni})$ is a map (a relator sends maps to maps), so it leaves Λ — $\triangleq$ relator, $Map(f)$ $\implies \frac{f\ R}{\ni} = f\ \frac{R}{\ni}$
5	$= \alpha^\circ\ F(\frac{X}{\ni})\ \frac{F(\ni)\ R}{\ni}$	fold uniqueness, $\frac{F(\ni)R}{\ni}$ being a map — $(h : f \to g) \triangleq f\ h = F(h)\ g$, $Map(f)$ $\implies \alpha\ (\!|f|\!) = F((\!|f|\!))\ f$
6	$= (\!|\frac{F(\ni)\ R}{\ni}|\!)$	

§5.6. Combinatorial functions

(5.6a)

	definition	type	note
1	`[A]::=nil|cons(A,[A])` `list(A) = [A]`	$\mathcal{A}\to\mathcal{A}$	The list type, under the short name it keeps.
2	`list(R) = ⦇[nil,(R × 𝟙) cons]⦈`	`[A]→[B]`	The relator's action on `R : A→B`: one `R` per element, the shape untouched.
3	`subseq = ⦇[nil,cons ∪ π₂]⦈`	`[A]→[A]`	`xs subseq ys`: `ys` is `xs` with elements dropped — `cons` keeps the head, `π₂` drops it.
4	`prefix = ⦇[nil,cons ∪ ⊸ nil]⦈` `prefix = cat° π₁ =init*`	`[A]→[A]`	`ys` is an initial segment of `xs`; the first `nil` is where it stops early. `init≜snoc° π₁`.
5	`suffix = cat° π₂ =tail*`	`[A]→[A]`	The dual, `tail≜cons° π₂`; as a reduce it needs snoc-lists.
6	`partition = concat°`	`[A]→[[A]]`	This `cat` is restricted to `[A]⁺×[A]→[A]`, so `ys` is a list of non-empty segments of `xs`.
7	`concat = ⦇[nil,cat]⦈`	`[[A]]→[A]`	Joins the segments back up, which is why its converse splits a list.
8	`inits`	`[A]→[[A]]`	Implements $\frac{\texttt{prefix}}{\ni}$, listing the prefixes by increasing length.
9	`tails`	`[A]→[[A]]`	Implements $\frac{\texttt{suffix}}{\ni}$ by decreasing length — the opposite order.

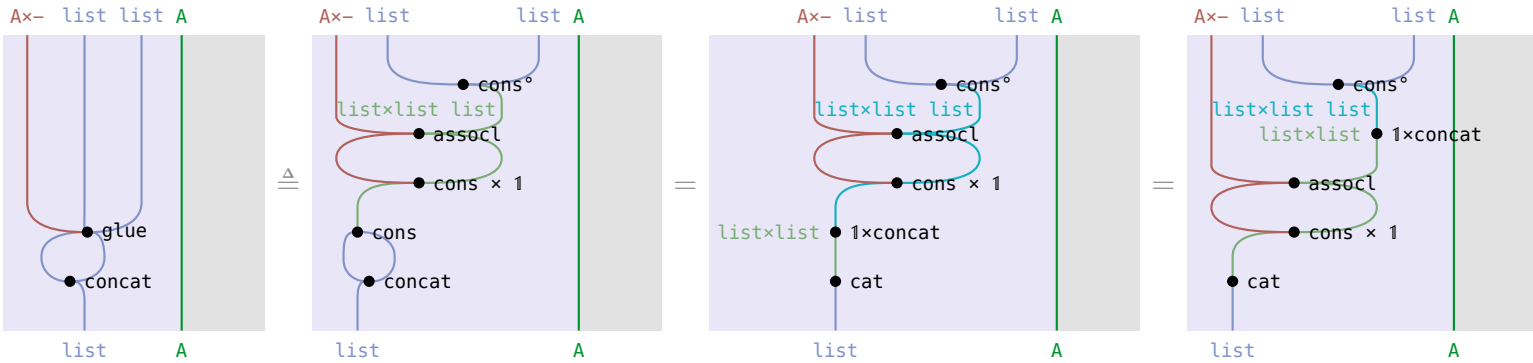
10	`cpp`, `cpl`, `cpr`	`[A]×[B]→[A×B]`
		`[A]×B→[A×B]`
		`A×[B]→[A×B]`
			Every pair `(a,b)` with `a` from the first list and `b` from the second; `cpl`, `cpr` keep one side a single element.

11	`cpp setify = (setify × setify)` $\frac{\ni\times\ni}{\ni}$ `cpl setify = (setify × 𝟙)` $\frac{\ni\times\mathbb{1}}{\ni}$ `cpr setify = (𝟙 × setify)` $\frac{\mathbb{1}\times\ni}{\ni}$	`[A]×[B]→P(A×B)`	Forgetting the order with `setify`, `cpp` is the transpose of `∋×∋`.
12	`cp(F)` $\triangleq \frac{\texttt{F}(\ni)}{\ni}$	`F(PA)→PFA`	`cp(F)`, the transpose of a relator's action on `∋`; `cpp` implements it at `F(A)=A×A`.
13	`cp(list) =` $\llparenthesis\frac{\texttt{[nil,}(\ni\times\ni)\texttt{ cons]}}{\ni}\rrparenthesis$ $\frac{\texttt{[nil,}(\ni\times\ni)\texttt{ cons]}}{\ni}$ `= [nil` $\frac{\mathbb{1}}{\ni}$, $\frac{\ni\times\ni}{\ni}$ `E(cons)]`	`list(PA)→P(list(A))`	`cp(list)`: (5.12) `⦇R⦈ =` $\llparenthesis\frac{\texttt{F}(\ni)\ \texttt{R}}{\ni}\rrparenthesis$ `∋` at `list(∋)=⦇[nil, (∋×𝟙)cons]⦈`, then the algebra expanded.
14	`cplist = ⦇[nil wrap,cpp list(cons)]⦈`	`[[A]]→[[A]]`	`cp(list)` with every set a list: `cpp` in place of $\frac{\ni\times\ni}{\ni}$.

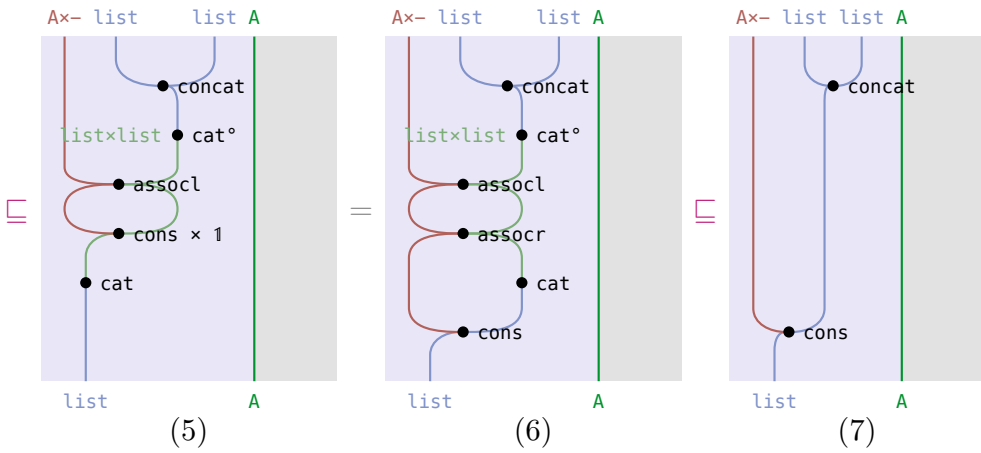
glue concat ⊑ (𝟙 × concat) cons (5.6b)

putting **a** on the front of the first segment and flattening (**glue concat**) gives at most flattening the segments and putting **a** on the front of the result (**(𝟙×concat) cons**)

the starting composite



1	glue concat	definition of glue
2	≜ (𝟙 × cons°) assocl cons × 𝟙 cons concat	since concat=⦇[nil,cat]⦈
3	= (𝟙 × cons°) assocl cons × 𝟙 (𝟙 × concat) cat	naturality of assocl
4	= (𝟙 × (cons° (𝟙 × concat))) assocl cons × 𝟙 cat	since concat=⦇[nil,cat]⦈



5	⊑ (𝟙 × concat) (𝟙 × cat°) assocl cons × 𝟙 cat	since (cons×𝟙) cat=assocr (𝟙×cat) cons
6	= (𝟙 × concat) (𝟙 × cat°) assocl assocr (𝟙 × cat) cons	since assocl assocr=𝟙 and cat° cat⊑𝟙
7	⊑ (𝟙 × concat) cons	

§5.6.1. $\frac{\text{subseq}}{\ni}$ = ⦇[nil $\frac{𝟙}{\ni}$,⟨$\frac{𝟙 \times \ni}{\ni}$ E(cons),π₂⟩ cup]⦈ — the set of subsequences is a fold of maps into sets

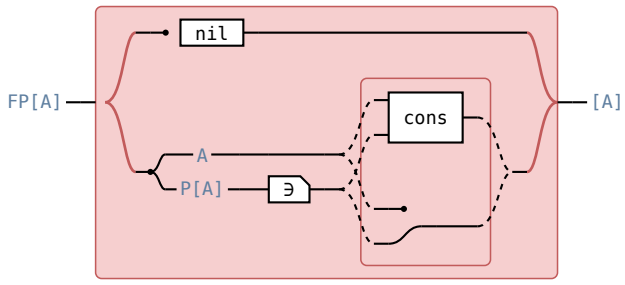
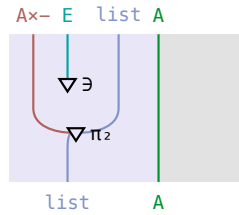
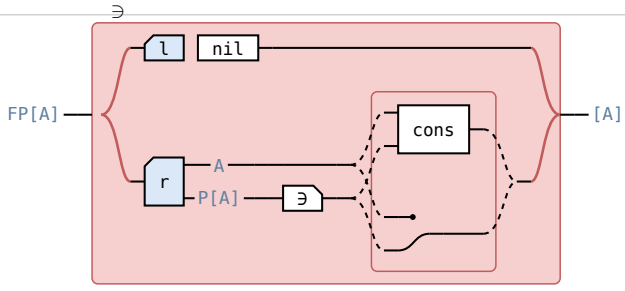
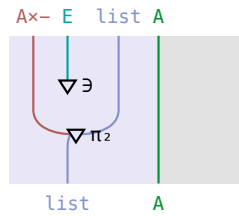
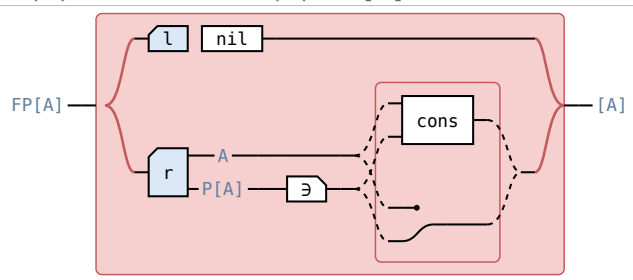
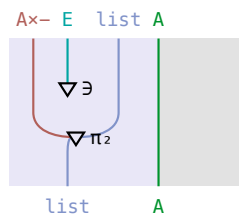
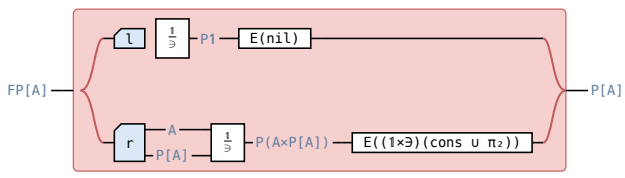
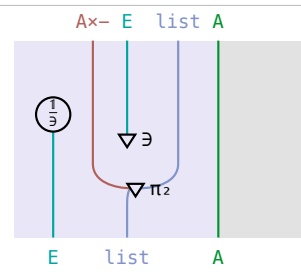
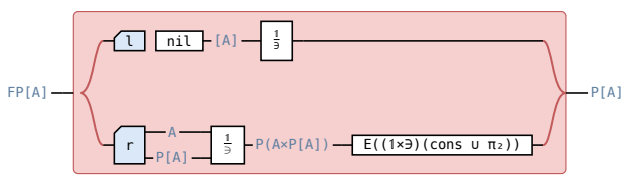
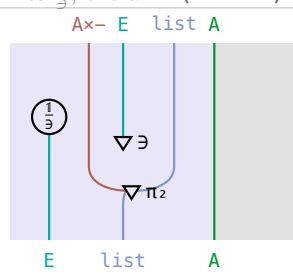
(5.6.1a)

cup ≜ $\frac{(\pi_1\ \ni)\ \cup\ (\pi_2\ \ni)}{\ni}$ — cup unions two sets.

cup≜ $\frac{\pi_1\ni\ \cup\ \pi_2\ni}{\ni}$: PA×PA→PA

$$\frac{\mathsf{F}(\ni)\ [\mathsf{nil},\mathsf{cons} \cup \pi_2]}{\ni} = [\mathsf{nil}\ \tfrac{\mathbb{1}}{\ni}, \frac{(\mathbb{1} \times \ni)\ (\mathsf{cons} \cup \pi_2)}{\ni}] \tag{5.6.1b}$$

the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12) $(\!|R|\!) = (\!|\frac{\mathsf{F}(\ni)\ R}{\ni}|\!)\ \ni$ at $\mathsf{subseq}=(\!|[\mathsf{nil},\mathsf{cons} \cup \pi_2]|\!)$, $\mathsf{subseq} = (\!|[\mathsf{nil},\mathsf{cons} \cup \pi_2]|\!)$.

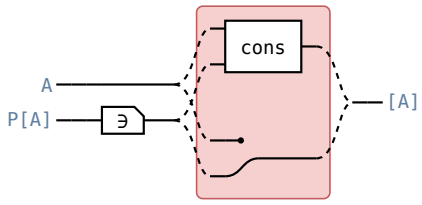
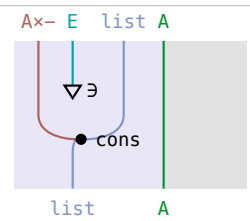
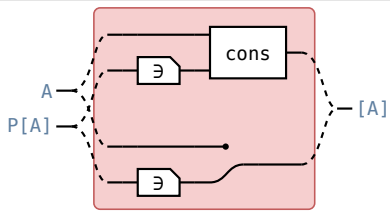
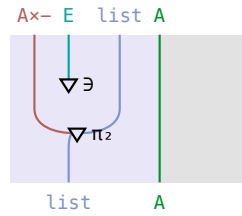
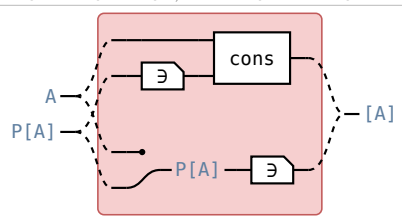
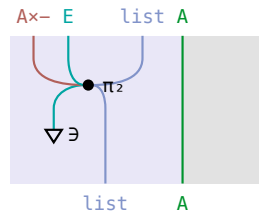
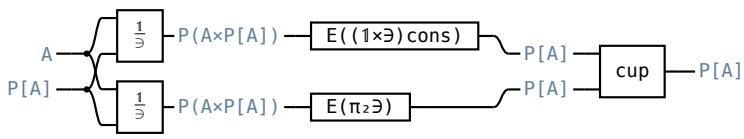
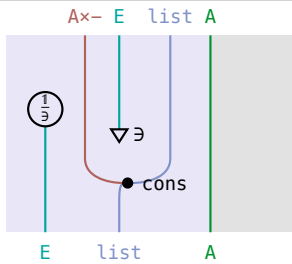
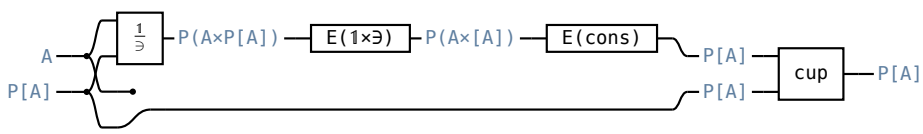
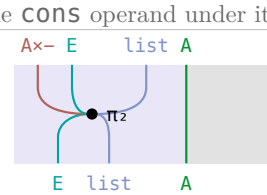
circuit — the fork is $\mathsf{F}([A])=\mathbb{1}+A\times[A]$: nil above, the pair below	**Hinze–Marsden**
 $\frac{\mathsf{F}(\ni)\ [\mathsf{nil},\mathsf{cons} \cup \pi_2]}{\ni}$	 the π_2 operand of $\mathsf{cons} \cup \pi_2$ under the $\mathbb{1}\times\ni$ summand of $\mathsf{F}(\ni)$, i.e. $(\mathbb{1}\times\ni)\pi_2$
$=$  $\frac{(\mathbb{1} + (\mathbb{1} \times \ni))\ [\mathsf{nil},\mathsf{cons} \cup \pi_2]}{\ni}$ $\mathsf{F}(X)=\mathbb{1}+A\times X$ — $\mathsf{list}(A) = [A]$	 the same operand under $\mathbb{1}+\mathbb{1}\times\ni$, whose $\mathbb{1}\times\ni$ summand it sits in
$=$  $\frac{[\mathsf{nil},(\mathbb{1} \times \ni)\ (\mathsf{cons} \cup \pi_2)]}{\ni}$ $R+S \triangleq [Rl,Sr]$, $l[R,S]=R$, $r[R,S]=S$ — $[R,S] \triangleq (l^\circ\ R) \cup (r^\circ\ S)$, $l\ [R,S] = R$, $r\ [R,S] = S$, $l\ T = R \wedge r\ T = S$ $\Longrightarrow T = [R,S]$	 the π_2 operand of the second arm $(\mathbb{1}\times\ni)(\mathsf{cons} \cup \pi_2)$
$=$  $[\frac{\mathsf{nil}}{\ni}, \frac{(\mathbb{1} \times \ni)\ (\mathsf{cons} \cup \pi_2)}{\ni}]$ $[R,S] = [\frac{R}{\ni},\frac{S}{\ni}]\ \ni$ at $T:=[\mathsf{nil},(\mathbb{1}\times\ni)(\mathsf{cons} \cup \pi_2)]$	 the π_2 operand under its $\frac{\mathbb{1}}{\ni}$, the arm $(\mathbb{1} \times \ni)\ \pi_2 = \pi_2\ \ni$ rewrites
$=$  $[\mathsf{nil}\ \tfrac{\mathbb{1}}{\ni}, \frac{(\mathbb{1} \times \ni)\ (\mathsf{cons} \cup \pi_2)}{\ni}]$ $\mathsf{Map}(f) \wedge \mathbb{1}\square = \ni\square$ $\Longrightarrow \frac{f}{\ni} = f\ \frac{\mathbb{1}}{\ni}$, $\frac{f}{\ni} = f\ \frac{\mathbb{1}}{\ni}$ for f a map, at $f:=\mathsf{nil}$	 the same operand; the two rows differ only in the nil arm

(5.6.1c)

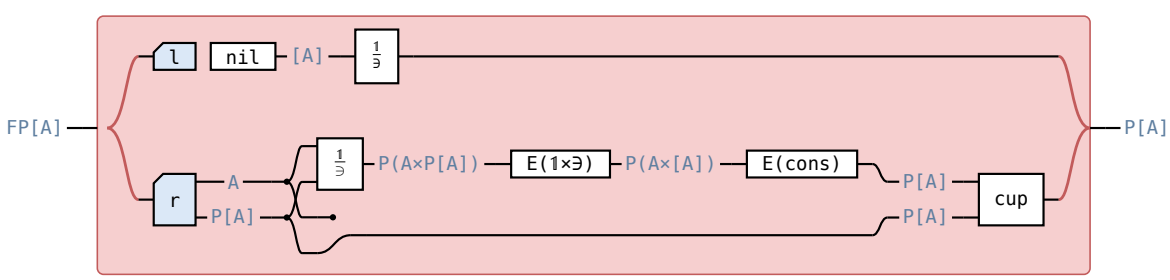
$$\begin{tikzcd}
A\times P[A] \arrow[r, "\pi_2"] \arrow[d, "\mathbb{1}\times\ni"'] & P[A] \arrow[d, "\ni"] \\
A\times[A] \arrow[r, "\pi_2"'] & {[A]}
\end{tikzcd}$$

$$\frac{(\mathbb{1} \times \ni)\ (\mathsf{cons} \cup \pi_2)}{\ni} = \langle \frac{\mathbb{1} \times \ni}{\ni}\ \mathsf{E}(\mathsf{cons}), \pi_2 \rangle\ \mathsf{cup} \tag{5.6.1d}$$

power transpose of join: the power transpose of the join of two relations is $\langle \frac{R}{\ni}, \frac{S}{\ni} \rangle$ cup, where cup is the function that returns the union of two sets

circuit	Hinze–Marsden
 $\frac{(\mathbb{1} \times \ni)\ (\mathsf{cons} \cup \pi_2)}{\ni}$ $\frac{\mathsf{F}(\ni)\ [\mathsf{nil}, \mathsf{cons} \cup \pi_2]}{\ni} = [\mathsf{nil}\ \frac{1}{\ni}, \frac{(\mathbb{1} \times \ni)\ (\mathsf{cons} \cup \pi_2)}{\ni}]$'s second branch	 the cons operand of $\mathsf{cons} \cup \pi_2$
=  $\frac{((\mathbb{1} \times \ni)\ \mathsf{cons}) \cup ((\mathbb{1} \times \ni)\ \pi_2)}{\ni}$ $\mathsf{T}(X_1 \cup X_2) = \mathsf{T}X_1 \cup \mathsf{T}X_2$ — $(S \cup T)\ R = (S\ R) \cup (T\ R)$, $R\ (S \cup T) = (R\ S) \cup (R\ T)$, $R \cap (S \cup T) = R \cap S \cup R \cap T$, $(R \cup S)^\circ = S^\circ \cup R^\circ$	 the π_2 operand of $(\mathbb{1} \times \ni)\mathsf{cons} \cup (\mathbb{1} \times \ni)\pi_2$
=  $\frac{((\mathbb{1} \times \ni)\ \mathsf{cons}) \cup (\pi_2\ \ni)}{\ni}$ $(\mathbb{1} \times \ni)\pi_2 = \pi_2 \ni$ — $(R \times S)\ \pi_1 \sqsubseteq \pi_1\ R \wedge (R \times S)\ \pi_2 \sqsubseteq \pi_2\ S$ at π_2, an equality because $\mathbb{1}$ is entire	 the π_2 operand of $(\mathbb{1} \times \ni)\mathsf{cons} \cup \pi_2 \ni$
=  $\langle \frac{(\mathbb{1} \times \ni)\ \mathsf{cons}}{\ni}, \frac{\pi_2\ \ni}{\ni} \rangle\ \mathsf{cup}$ $\frac{R \cup S}{\ni} = \langle \frac{R}{\ni}, \frac{S}{\ni} \rangle\ \mathsf{cup}$ — $\triangleq$ cup	 the cons operand under its $\frac{1}{\ni}$
=  $\langle \frac{\mathbb{1} \times \ni}{\ni}\ \mathsf{E}(\mathsf{cons}), \pi_2 \rangle\ \mathsf{cup}$ $\frac{S}{\ni}\ \mathsf{E}(R) = \frac{S\ R}{\ni}$, absorption $\frac{S}{\ni}\ \mathsf{E}(R) = \frac{SR}{\ni}$ at $S := \mathbb{1} \times \ni$, $R := \mathsf{cons}$; fusion and $\frac{\ni}{\ni} = \mathbb{1}$ on the π_2 operand	 the π_2 operand, bare π_2

(5.6.1e)



$$[\frac{\mathsf{nil}}{\ni}, \langle \frac{\mathbb{1} \times \ni}{\ni}\ \mathsf{E}(\mathsf{cons}), \pi_2 \rangle\ \mathsf{cup}]$$

which writes Pcons; cons is a map, and there $\mathsf{P}(\mathsf{cons}) = \mathsf{E}(\mathsf{cons})$ — $\mathsf{Map}(f)$

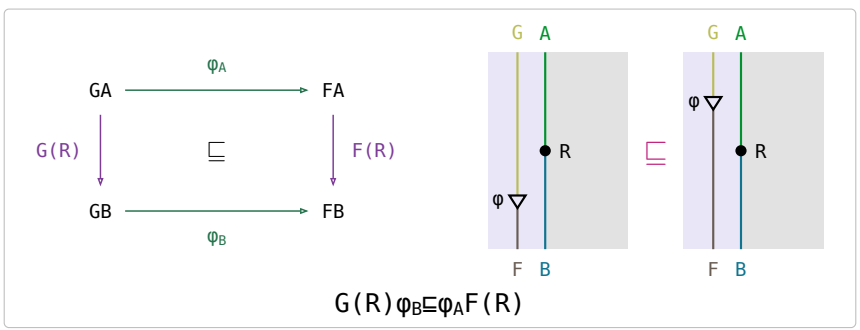
$\Longrightarrow \mathsf{P}(f) = \mathsf{E}(f)$.

§5.7. Lax natural transformations (LaT)

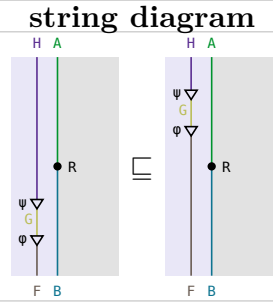
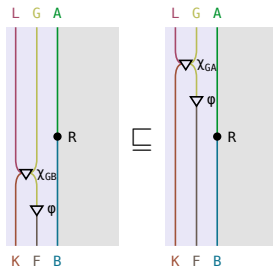
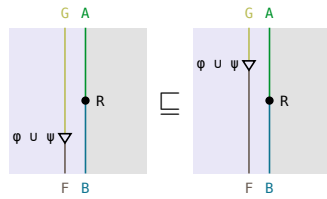
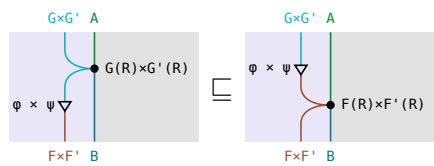
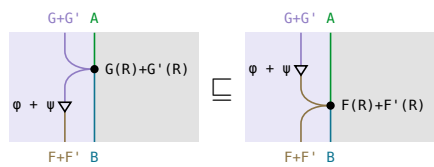
(5.7a)

$\varphi : G \Rightarrow F$ is **lax at** $R : A \to B$ when $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$; it is a **lax natural transformation** (LaT) when lax at every R.

(5.7b)



(5.7c)

	closed under	commutative diagram	string diagram
1	composition $\psi\varphi$	$\begin{tikzcd} HA \arrow[r, "\psi_A"] \arrow[d, "H(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & GA \arrow[r, "\varphi_A"] \arrow[d, "G(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA \arrow[d, "F(R)"] \\ HB \arrow[r, "\psi_B"'] & GB \arrow[r, "\varphi_B"'] & FB \end{tikzcd}$ $H(R)\psi_B \sqsubseteq \psi_A G(R)$ and $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ give $H(R)(\psi_B\varphi_B) \sqsubseteq (\psi_A\varphi_A)F(R)$	
2	horizontal composition $\chi\circ\varphi$	$\begin{tikzcd} LGA \arrow[r, "\chi_{GA}K(\varphi_A)"] \arrow[d, "L(G(R))"'] \arrow[dr, phantom, "\sqsubseteq"] & KFA \arrow[d, "K(F(R))"] \\ LGB \arrow[r, "\chi_{GB}K(\varphi_B)"'] & KFB \end{tikzcd}$ $\varphi : G \Rightarrow F$ with $G,F : \mathcal{C}\to\mathcal{D}$ and $\chi : L \Rightarrow K$ with $L,K : \mathcal{D}\to\mathcal{E}$ give $\chi\circ\varphi : L\circ G \Rightarrow K\circ F$, the family $A \mapsto \chi_{GA}K(\varphi_A)$ $L(G(R))\chi_{GB} \sqsubseteq \chi_{GA}K(G(R))$ is χ lax at $G(R)$; then $K(G(R)\varphi_B) \sqsubseteq K(\varphi_A F(R))$ is K applied to φ's own inequation; $\chi := \mathbb{1}_K$ gives $K(\varphi) : K\circ G \Rightarrow K\circ F$, K composed on the outside; $\varphi := \mathbb{1}_G$ gives $\chi G : L\circ G \Rightarrow K\circ G$, G composed on the inside two candidates, $\chi_{GA}K(\varphi_A)$ and $L(\varphi_A)\chi_{FA}$, ordered by $\sqsubseteq$; this row is the first	
3	union $\varphi \cup \psi$	$\begin{tikzcd} GA \arrow[r, "\varphi_A"] \arrow[d, "G(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA \arrow[d, "F(R)"] \\ GB \arrow[r, "\varphi_B"'] & FB \end{tikzcd} \cup \begin{tikzcd} GA \arrow[r, "\psi_A"] \arrow[d, "G(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA \arrow[d, "F(R)"] \\ GB \arrow[r, "\psi_B"'] & FB \end{tikzcd}$ $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ and $G(R)\psi_B \sqsubseteq \psi_A F(R)$ give $G(R)(\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A)F(R)$	
4	a relator K $K(\varphi)$	$\begin{tikzcd} KGA \arrow[r, "K(G(R))"] \arrow[d, "K(\varphi_A)"'] \arrow[dr, phantom, "\sqsubseteq"] & KGB \arrow[d, "K(\varphi_B)"] \\ KFA \arrow[r, "K(F(R))"'] & KFB \end{tikzcd}$ $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ gives $K(G(R))K(\varphi_B) \sqsubseteq K(\varphi_A)K(F(R))$	
5	product $\varphi\times\psi$	$\begin{tikzcd} GA\times G'A \arrow[r, "\varphi_A\times\psi_A"] \arrow[d, "G(R)\times G'(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA\times F'A \arrow[d, "F(R)\times F'(R)"] \\ GB\times G'B \arrow[r, "\varphi_B\times\psi_B"'] & FB\times F'B \end{tikzcd}$ $\varphi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\varphi\times\psi : G\times G' \Rightarrow F\times F'$ $(R\times S)(U\times V) = (RU)\times(SV)$ and monotonicity in both slots — the row above's K applied to the inequation, run on a bifunctor; the fork is the derived case $\langle\varphi,\psi\rangle = \triangleleft(\varphi\times\psi)$, and its ONE cost is the lax copy law $R\triangleleft \sqsubseteq \triangleleft(R\times R)$ — $R \triangleleft \leq \triangleleft (R \otimes R)$, $R \multimap \leq \multimap$	
6	coproduct $\varphi+\psi$	$\begin{tikzcd} GA+G'A \arrow[r, "\varphi_A+\psi_A"] \arrow[d, "G(R)+G'(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA+F'A \arrow[d, "F(R)+F'(R)"] \\ GB+G'B \arrow[r, "\varphi_B+\psi_B"'] & FB+F'B \end{tikzcd}$ $\varphi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\varphi+\psi : G+G' \Rightarrow F+F'$ $(R+S)(U+V) = (RU)+(SV)$ and monotonicity in both slots; the cofork is the derived case $[\varphi,\psi] = (\varphi+\psi)\triangledown$, and $\triangledown$ costs nothing	
7	meet — **fails** $\varphi\cap\psi$	the step it would need is $(\varphi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\varphi\cap\psi)_A F(R)$, the wrong direction of $R\ (S \cap T) \sqsubseteq (R\ S) \cap (R\ T)$ a counterexample: $aa = (0,1) \wedge bb = (0,0) \wedge (R \times R)(aa,bb) \wedge b_1 = \{0\} \wedge b_1 = E(\pi_1 \cap \pi_2)\{bb\} \wedge a_1 = \varnothing \wedge a_1 = E(\pi_1 \cap \pi_2)\{aa\} \wedge b_2 = \varnothing \wedge b_2 = E(R)(a_1)$ $\Longrightarrow \neg((R \times R)\ (\pi_1 \cap \pi_2) \sqsubseteq (\pi_1 \cap \pi_2)\ R)$	
8	Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A(XY) \sqsubseteq (X'Y')R_C$ at $R_A,R_B := G(R),F(R)$ and $X,X' := \varphi_B,\varphi_A$, quantified over R; at $X'=X$ with R_A,R_B endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference		

§5.7.1. meet is not closed in LaT

(5.7.1a)

counterexample — φ,ψ : G⇒F lax natural does NOT give φ∩ψ lax natural

$$\begin{tikzcd}
\begin{matrix} A\times A \\ (0,1) \end{matrix} \arrow[r, "\pi_1\cap\pi_2"] \arrow[d, "R\times R"'] \arrow[dr, phantom, "\not\sqsubseteq"] & \begin{matrix} A \\ \varnothing \end{matrix} \arrow[d, "R"] \\
\begin{matrix} B\times B \\ (0,0) \end{matrix} \arrow[r, "\pi_1\cap\pi_2"'] & \begin{matrix} B \\ \{0\} \\ \varnothing \end{matrix}
\end{tikzcd}$$

$A=B\triangleq\{0,1\}$, $R\triangleq\{(0,0),(1,0)\}$, $\varphi\triangleq\pi_1\cap\pi_2 : \Delta\Rightarrow Id$

§5.7.2. Two stacked towers

(5.7.2a)

		lower — inside ONE allegory	upper — between allegories
1	0-cell	an object A	an allegory $\mathcal{C}$ — **a whole lower tower**
2	1-cell	a relation R : A→B	a relator F : $\mathcal{C}\to\mathcal{D}$
3	2-cell	R⊑S	a LaT φ : G⇒F ≜ lax
4	order	R⊑S	φ⊑ψ componentwise
5	union	R ∪ S	φ ∪ ψ, **survives** lax(φ) ∧ lax(ψ) ⟹ G(R) $(\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A)$ F(R)
6	product	R×S ≜ ×	φ×ψ, **survives**; a TENSOR, not a categorical product; SYMMETRIC ⟨R,S⟩ π_1 = dom(S) R , ⟨R,S⟩ π_2 = dom(R) S lax(φ) ∧ lax(ψ) ⟹ (G(R) × G'(R)) $(\varphi_B \times \psi_B) \sqsubseteq (\varphi_A \times \psi_A)$ (F(R) × F'(R))
7	coproduct	R+S	φ+ψ, **survives**; a BIPRODUCT lax(φ) ∧ lax(ψ) ⟹ (G(R) + G'(R)) $(\varphi_B + \psi_B) \sqsubseteq (\varphi_A + \psi_A)$ (F(R) + F'(R))
8	meet	R∩S	φ∩ψ componentwise, **fails** aa = (0,1) ∧ bb = (0,0) ∧ (R × R)(aa,bb) ∧ b_1 = {0} ∧ b_1 = E(π_1 ∩ π_2){bb} ∧ a_1 = ∅ ∧ a_1 = E(π_1 ∩ π_2){aa} ∧ b_2 = ∅ ∧ b_2 = E(R)(a_1) ⟹ ¬((R × R) (π_1 ∩ π_2) ⊑ (π_1 ∩ π_2) R)
9	converse	R°	φ°, **fails**, oplax: φ_A°G(R)⊑F(R)φ_B°
10	zero object	z with 𝟙 z=𝟘	the constant relator at such a z, initial and terminal

§5.7.3. R⊑S is a special case of LaT

(5.7.3a)

$$\begin{tikzcd}
GA \arrow[r, "\varphi_A"] \arrow[d, "G(R)"'] \arrow[dr, phantom, "\sqsubseteq"] & FA \arrow[d, "F(R)"] \\
GB \arrow[r, "\varphi_B"'] & FB
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
A \arrow[r, "\varphi_X"] \arrow[d, "\mathbb{1}"'] \arrow[dr, phantom, "\sqsubseteq"] & B \arrow[d, "\mathbb{1}"] \\
A \arrow[r, "\varphi_Y"'] & B
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
A \arrow[r, bend left, "\varphi_Y"] \arrow[r, bend right, "\varphi_X"'] \arrow[r, phantom, "\sqsubseteq"] & B
\end{tikzcd}$$

G≜const A, F≜const B — the relators X↦A and X↦B, each sending every arrow to 𝟙 ≜ lax

with $\mathcal{C}$ two objects and one non-identity arrow X→Y, a LaT const A⇒const B is the pair φ_X,φ_Y : A→B with $\varphi_Y\sqsubseteq\varphi_X$ — the datum R⊑S

not every LaT is constant: ∋ : P⇒Id X ⊑ P(R) ⟺ X ∋ ⊑ ∋ R ∧ X° ∋ ⊑ ∋ R° π_2 : (A×−)⇒Id (𝟙 × ∋) π_2 = π_2 ∋

◁ : Id⇒Δ R ◁ ≤ ◁ (R ⊗ R) , R ⊸ ≤ ⊸